

Exercise 2.12

- (a) Construct a table of p_x for Makeham's law with parameters $A = 0.0001$, $B = 0.00035$, and $c = 1.075$, for integer x from age 0 to age 130, using Excel or another appropriate computer software. You should set the parameters so that they can be easily changed, and you should keep the table, as many exercises and examples in future chapters will use Makeham's law.
- (b) Use the table to determine the age last birthday at which a life currently aged 70 is most likely to die.
- (c) Use the table to calculate e_{70} .
- (d) Using a numerical approach, calculate ${}^{\circ}e_{70}$.

Solution

For Makeham's law,

$$\mu_x = A + Bc^x.$$

Therefore the one-year survival probability is

$$p_x = {}_1p_x = \exp \left\{ - \int_x^{x+1} (A + Bc^r) dr \right\} = \exp \left\{ -A - \frac{B}{\log c} c^x (c - 1) \right\}.$$

With

$$A = 0.0001, \quad B = 0.00035, \quad c = 1.075,$$

this gives the following table.

x	p_x	x	p_x	x	p_x	x	p_x
0	0.999537	33	0.995960	66	0.957877	99	0.626825
1	0.999510	34	0.995666	67	0.954797	100	0.605252
2	0.999481	35	0.995349	68	0.951497	101	0.582887
3	0.999449	36	0.995008	69	0.947963	102	0.559766
4	0.999415	37	0.994642	70	0.944178	103	0.535933
5	0.999379	38	0.994249	71	0.940127	104	0.511442
6	0.999340	39	0.993827	72	0.935790	105	0.486362
7	0.999298	40	0.993373	73	0.931151	106	0.460771
8	0.999253	41	0.992885	74	0.926190	107	0.434760
9	0.999204	42	0.992361	75	0.920886	108	0.408434
10	0.999152	43	0.991797	76	0.915218	109	0.381908
11	0.999096	44	0.991192	77	0.909164	110	0.355311
12	0.999036	45	0.990542	78	0.902700	111	0.328782
13	0.998971	46	0.989844	79	0.895803	112	0.302468
14	0.998902	47	0.989094	80	0.888447	113	0.276525
15	0.998827	48	0.988288	81	0.880607	114	0.251112
16	0.998746	49	0.987423	82	0.872256	115	0.226392
17	0.998660	50	0.986493	83	0.863368	116	0.202525
18	0.998567	51	0.985495	84	0.853913	117	0.179666
19	0.998467	52	0.984423	85	0.843865	118	0.157963
20	0.998360	53	0.983272	86	0.833195	119	0.137547
21	0.998244	54	0.982036	87	0.821875	120	0.118532
22	0.998120	55	0.980709	88	0.809878	121	0.101013
23	0.997987	56	0.979285	89	0.797176	122	0.085057
24	0.997843	57	0.977756	90	0.783744	123	0.070704
25	0.997689	58	0.976115	91	0.769557	124	0.057963
26	0.997524	59	0.974354	92	0.754591	125	0.046816
27	0.997346	60	0.972465	93	0.738828	126	0.037211
28	0.997154	61	0.970438	94	0.722250	127	0.029072
29	0.996949	62	0.968264	95	0.704843	128	0.022297
30	0.996728	63	0.965932	96	0.686598	129	0.016764
31	0.996490	64	0.963431	97	0.667511	130	0.012337
32	0.996235	65	0.960750	98	0.647584		

(b)

According to the constructed life table, the probability of dying in each specific year is given by the deferred mortality probability

$${}_t p_{70} q_{70+t}.$$

For example, near the maximum,

$${}_2 p_{70} q_{72} \approx 0.056733, \quad {}_3 p_{70} q_{73} \approx 0.056643.$$

The largest value occurs at the interval associated with age 72. Since the policyholder completes the 72nd year and enters the subsequent risk period, a death occurring in this maximum-probability interval means that the individual will have celebrated the 73rd birthday before the end of the full period. Therefore, the age last birthday is 73.

(c)

The curtate expectation of life is

$$e_{70} = \sum_{k=1}^{\infty} {}_k p_{70}.$$

Using the table recursively, with

$${}_k p_{70} = p_{70} p_{71} \cdots p_{70+k-1},$$

gives

$$e_{70} \approx 9.339.$$

(d)

The complete expectation of life is

$${}^{\circ}e_{70} = \int_0^{\infty} {}_t p_{70} dt.$$

For Makeham's law,

$${}_t p_{70} = \exp \left\{ -At - \frac{B}{\log c} c^{70} (c^t - 1) \right\}.$$

Hence a numerical integration gives

$${}^{\circ}e_{70} = \int_0^{\infty} \exp \left\{ -0.0001t - \frac{0.00035}{\log(1.075)} (1.075)^{70} ((1.075)^t - 1) \right\} dt \approx 9.834.$$